

**ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)**  
**ORGANISATION OF ISLAMIC COOPERATION (OIC)**  
**Department of Computer Science and Engineering (CSE)**

**MID SEMESTER EXAMINATION**  
**DURATION: 1 HOUR 30 MINUTES**

**WINTER SEMESTER, 2024-2025**  
**FULL MARKS: 75**

**CSE 4539: Web Programming**

**Programmable calculators are not allowed. Do not write anything on the question paper.**  
**Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.**

1. a) Implement the CSS to achieve the following outcome for Figure 1.

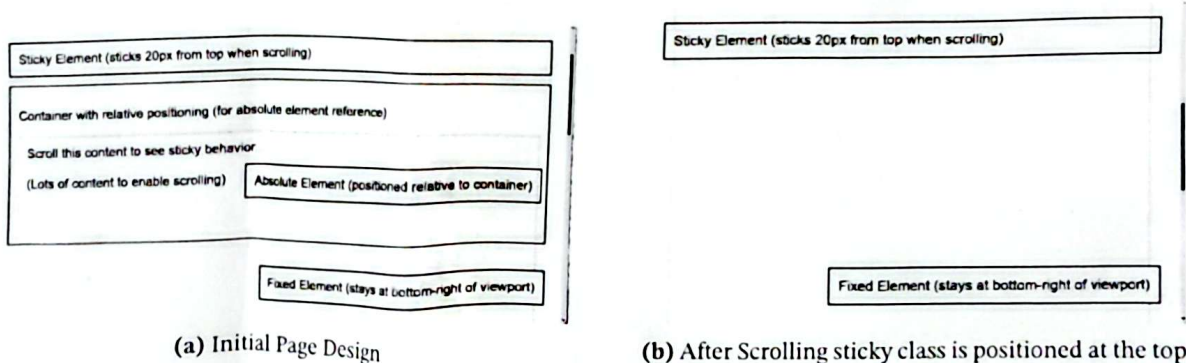
3 × 5  
(CO2)  
(PO2)

```

1 <body>
2   <div class="sticky">Sticky Element (sticks 20px from top when
3     scrolling)</div>
4
5   <div class="container">
6     <p>Container with relative positioning (for absolute element
7       reference)</p>
8     <div class="absolute">Absolute Element (positioned 50px top
9       relative to container)</div>
10    <div class="content">
11      Scroll this content to see sticky behavior...
12      <br><br>(Lots of content to enable scrolling)
13    </div>
14  </div>
15
16  <div class="fixed">Fixed Element (stays at bottom-right of
17    viewport)</div>
18 </body>
  
```

**Code Snippet 1: HTML with CSS Positioning Example**

The output of Code Snippet 1 is given bellow:



**Figure 1: Output for the given Code Snippet 1**

- b) Perform a following DOM manipulation using a scripting language-
- Change the `innerHTML` of `sticky` class to - "This is an example of Sticky position."
  - Add a `h1` heading at the top of the page titled- "Different CSS Positions".

4 × 2.5  
(CO2)  
(PO1)

- iii. Replace the newly created h1 tag with a h3 tag titled- "CSS Positions: Absolute, Fixed and Sticky".
  - iv. Add an event handler to fixed class that will show an alert message- "This is an example of Fixed position".
2. You are building a simple backend system for "No Pain No Lie Gym" where honesty is mandatory. Each gym member has basic information such as id, name, age, membershipType, and weight. The database connection and required schemas are already configured.

5 × 5  
(CO2)  
(PO2)

Design appropriate API routes with proper response codes for the following operations using javascript:

- a) A new member is saved only if their age is greater than 18. Anyone younger is politely rejected with a suggestions of joining a "Yoga for Toddlers" program.
  - b) Members sorted by age (youngest to oldest) to decide who gets cardio and who gets "light stretching".
  - c) When a member disputes their membership, the system retrieves their record using a specific member ID.
  - d) The gym wants to run targeted motivational campaigns for members with Yearly membership and weight over 80 kg.
  - e) If a members information changes (e.g., weight), only the modified field is updated, leaving the rest unchanged.
3. You are given a professional training workshop schedule in Table 1

Table 1: Professional Training Workshop Schedule for Question 3

| Date       | Schedule |          |      |             |                 |
|------------|----------|----------|------|-------------|-----------------|
|            | Schedule |          | Room | Trainer     | Workshop Title  |
|            | Begin    | End      |      |             |                 |
| 15.01.2024 | 9:00 AM  | 11:00 AM | 201  | Dr. Rahman  | Data Analytics  |
|            | 12:00 PM | 2:00 PM  | 305  | Ms. Akter   | UI/UX Design    |
| 16.01.2024 | 10:00 AM | 12:30 PM | 210  | Mr. Khan    | Cloud Computing |
|            | 2:00 PM  | 4:00 PM  | 410  | Dr. Ahmed   | Cybersecurity   |
| 17.01.2024 | 9:30 AM  | 12:00 PM | 108  | Ms. Sultana | Agile Project   |

- a) With proper class, id and name write the corresponding HTML code (only the table part) for Table 1.
- b) Style Table 1 with CSS having the following requirements:
  - i. Change the background color of alternating table rows.
  - ii. Align the text in the "Trainer" column to the left while keeping other columns centered.
  - iii. Table headers are distinguishable from table rows.
  - iv. Add a hover effect that highlights the entire row when the user moves the mouse over it.
  - v. Add a subtle outer border with rounded corners around the entire table.

10  
(CO2)  
(PO1)

5 × 3  
(CO2)  
(PO2)